

Data driven approach via discrete rough paths

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Outline

Motivation examples

- Gradient decent method

- Neural differential equations

Rough path theory

Discrete rough paths

- Rough difference equations

Applications

Gradient decent method

- Minimization problem

$$\min_{\theta \in \mathbb{R}^d} Q(\theta), \quad Q \in C^1. \quad (1.1)$$

- Global solution: $\theta^* \in \arg \min_{\theta \in \mathbb{R}^d} Q(\theta) \Rightarrow \nabla Q(\theta^*) = 0$. This leads to an approximation scheme to achieve θ^* , which is the zero solution of $\nabla Q(\theta) = 0$

$$\theta_{t+1} = \theta_t - \eta \nabla Q(\theta_t), \quad (1.2)$$

for the learning rate $\eta > 0$. (1.2) is called Gradient descent method.

- When η small enough, (1.2) can also be viewed as the Euler scheme for solving the ODE

$$\dot{\theta} = -\nabla Q(\theta), \quad (1.3)$$

where one attempts to solve for the steady state θ^* of (1.3).

Stochastic gradient descent

- Minimization problem

$$\min_{\theta \in \mathbb{R}^d} Q(\theta) = \min_{\theta \in \mathbb{R}^d} \frac{1}{m} \sum_{i=1}^m Q^{(i)}(\theta), \quad Q^{(i)} \in \mathcal{C}^1. \quad (1.4)$$

When $m \gg 1$, (1.2) is replaced by: at each step t choose a random set Ω_t of $\{1, \dots, m\}$ (with or without replacement) and then updating

$$\theta_{t+1} := \theta_t - \eta \frac{1}{|\Omega_t|} \sum_{i \in \Omega_t} \nabla Q^{(i)}(\theta_t) = \theta_t - \eta \nabla \tilde{Q}_t(\theta_t) \quad (1.5)$$

No matter how random the choice, can rewrite SGD recursion

$$\theta_{t+1} = \theta_t - \eta \nabla Q(\theta_t) + \eta^\alpha \epsilon_t(\theta_t). \quad (1.6)$$

- (1.6) can also be viewed as the Euler scheme of the "limiting equation"

$$d\theta = -\nabla Q(\theta)dt + \sigma(\theta)dX_t, \quad (1.7)$$

where $\sigma\sigma^T = \eta^{2\alpha} \mathbb{E}[\epsilon\epsilon^T | \theta_t = \theta]$.

- If Ω_t i.i.d. then CLT gives $\alpha = \frac{1}{2}$ and X -Gaussian. But not necessary i.i.d.
- **Questions:** How does the distribution of (1.6) converge to some limit?
And which calculus is used to interpret and solve the limiting equation?

Residual Neural Networks via rough paths

- Plain neural network: hidden layer equation

$$h_0 = x, \quad h_{t+1} = f(W_t h_t + b_t), \quad y = L(h_{t+1})$$

In the ResNet approach,

$$h_{t+1} = h_t + f(W_t h_t + b_t)\eta. \quad (1.8)$$

- Chen - Rubanova - Bettencourt - Duvenaud (2018): if one iterates continuously with infinitely many layers, one can view (1.8) as a step of the Euler scheme of a controlled ODE - neural differential equation,

$$\dot{h}_t = f_\theta(h_t + b_t), \quad h_0 = x. \quad (1.9)$$

Thus

$$L(h_T) = L(x + \int_0^T f_\theta(h_t + b_t))$$

- Problem: for time series data, the scheme is very unstable every time a new data is included, making a jump.

Residual Neural Networks via rough paths

- Kidger - Chen - Lyons (2021): view it as a controlled differential equation

$$h_{t+1} = h_t + f_\theta(h_t + b_t)dX_t \Rightarrow dh_t = f_\theta(h_t + b_t)dX_t \quad (1.10)$$

- Stability of ResNet (Bayer-Friz-Tapia, 2023):

$$\begin{aligned} \sup_{k=0..n} \|h_k - \tilde{h}_k\| &\leq C'_{p,n} \exp \left\{ C_{p,n} \|f\|_{C_b^3}^p \left(\|\mathbf{x}\|_{p,[0,n]}^p + \|\tilde{\mathbf{x}}\|_{p,[0,n]}^p \right) \right\} \\ &\quad \times (\|\mathbf{x}_0 - \tilde{\mathbf{x}}_0\| + \rho_p(\mathbf{x}, \tilde{\mathbf{x}})) \end{aligned} \quad (1.11)$$

Signatures of rough paths

Let $x \in C^{1-var}(I, \mathbb{R}^d)$. x_t^i is the coordinate i of x_t at time t . Define

$$\mathbf{g}^{k; i_1, \dots, i_k} := \int_s^t \int_s^{u_k} \dots \int_s^{u_2} dx_{u_1}^{i_1} \dots dx_{u_k}^{i_k}.$$

and the set

$$\mathbf{g} = \left(\mathbf{g}^{k; i_1, \dots, i_k} : 1 \leq k \leq N; i_1, \dots, i_k \in \{1, \dots, d\} \right)$$

to be the N -level signature of $x|_{[s,t]}$ and is written as $S_N(x)_{s,t}$. In other words,

$$S_N(x)_{s,t} \equiv \left(1, \int_{s < u < t} dx_u, \dots, \int_{s < u_1 < \dots < u_k < t} dx_{u_1} \otimes \dots \otimes dx_{u_k} \right)$$

Then

$$\in \oplus_{k=0}^N \left(\mathbb{R}^d \right)^{\otimes k},$$

$S_N(x)_{s, \cdot}$ satisfies the controlled DE by x as follows

$$\begin{cases} dS_N(x)_{s,t} = S_N(x)_{s,t} \otimes dx_t, \\ S_N(x)_{s,s} = 1. \end{cases}$$

Rough differential equation

$$dy = V(y)dx \Rightarrow y_{s,t} \approx \tilde{V}(y_s) \otimes S_N(x)_{s,t}$$

Example: $dy = V(y)dB^H$. Ito calculus provides only measurability of $y(T, B^H(\omega), y_0)$. While rough path theory proves continuous dependence (Lyon's universal theorem).

Rough paths

In the simplest form for $\beta \in (\frac{1}{3}, \frac{1}{2})$, a couple $\mathbf{x} = (x, \mathbb{X})$, with $x \in C^\beta([a, b], \mathbb{R}^d)$ and $\mathbb{X} \in C^{2\beta}_2([a, b]^2, \mathbb{R}^d \otimes \mathbb{R}^d)$ is called a *rough path* if they satisfy Chen's relation

$$\mathbb{X}_{s,t} - \mathbb{X}_{s,u} - \mathbb{X}_{u,t} = x_{s,u} \otimes x_{u,t}, \quad \forall a \leq s \leq u \leq t \leq b. \quad (2.1)$$

We write $\mathbb{X}_{s,t} = (x \otimes x)_{s,t} =: \int_s^t x_{s,u} \otimes dx_u$. In a general form, $(y \otimes x)_{s,t} =: \int_s^t y_{s,u} \otimes dx_u$ satisfies Chen's relation

$$(y \otimes x)_{s,t} - (y \otimes x)_{s,u} - (y \otimes x)_{u,t} = y_{s,u} \otimes x_{u,t}.$$

Denote by $\mathcal{C}^\beta([a, b]) \subset C^\beta \oplus C^{2\beta}$ the set of all rough paths in $[a, b]$, then $\mathcal{C}^\beta$ is a closed set but not a linear space, equipped with the rough path semi-norm

$$\|\mathbf{x}\|_{\beta, [a, b]} := \|x\|_{\beta, [a, b]} + \|\mathbb{X}\|_{2\beta, [a, b]^2}^{\frac{1}{2}} < \infty. \quad (2.2)$$

Denote by $T^2(\mathbb{R}^d) = \mathbb{R} \oplus \mathbb{R}^d \oplus \mathbb{R}^{d \otimes 2}$ the truncated step-2 tensor algebra, then $T^2(\mathbb{R}^d)$ is a Banach space with the norm $|\omega| := \max_{k=0,1,2} \|\pi_k(\omega)\|$ where $\|\pi_k(\omega)\|$ is Euclidean norm on $(\mathbb{R}^d)^{\otimes k}$. Define $T_1^2(\mathbb{R}^d) = \{\omega \in T^2(\mathbb{R}^d) : \pi_0(\omega) = 1\}$ together with a $\bullet$ operator

$$(1, g^1, g^2) \bullet (1, h^1, h^2) = (1, g^1 + h^1, g^2 + h^2 + g^1 \otimes h^1), \quad (2.3)$$

for any $\mathbf{g} = (1, g^1, g^2)$, $\mathbf{h} = (1, h^1, h^2)$. Then $(T_1^2(\mathbb{R}^d), \bullet)$ is a group with the identity element $\mathbf{1} = (1, 0, 0)$, and each element $\mathbf{g}$ has its inverse element $\mathbf{g}^{-1} = (1, -g^1, -g^2 - g^1 \otimes g^1)$. In addition, for a rough path $\mathbf{x}$ one can easily check from Chen's relation (??) that

$$(1, \mathbf{x}_{s,u}, \mathbb{X}_{s,u}) \bullet (1, \mathbf{x}_{u,t}, \mathbb{X}_{u,t}) = (1, \mathbf{x}_{s,t}, \mathbb{X}_{s,t}), \quad \forall s \leq u \leq t \in \Pi.$$

Also, one can view $\mathbf{x}_t = (1, \mathbf{x}_{0,t}, \mathbb{X}_{0,t})$ as a map from Π to $T_1^2(\mathbb{R}^d)$.

Rough difference equations

Discrete time set $\Pi = \{t_k\}_{k \in \mathbb{N}}$ defined by

$$y_{t_0} \in \mathbb{R}^d, \quad y_{t_{k+1}} = y_{t_k} + f(y_{t_k})(t_{k+1} - t_k) + F_{t_k, t_{k+1}}(y, \mathbf{x}), \quad (3.1)$$

$$F_{s,t}(y, \mathbf{x}) = \begin{cases} g(y_s)x_{s,t}, & \mathbf{x} = x \\ g(y_s)x_{s,t} + Dg(y_s)g(y_s)\mathbb{X}_{s,t}, & \mathbf{x} = (x, \mathbb{X}) \end{cases} \quad (3.2)$$

(H_f) $f \in C(\mathbb{R}^d, \mathbb{R}^d)$ of linear growth, i.e.

$$\exists C_f > 0 : \quad \|f(y)\| \leq C_f \|y\| + \|f(0)\|, \quad \forall y \in \mathbb{R}^d; \quad (3.3)$$

(H_g) either $g \in C_b^2(\mathbb{R}^d, \mathbb{R}^{d \times m})$ such that

$$\|g\|_\infty, C_g := \max \left\{ \|Dg\|_\infty, \|D^2g\|_\infty \right\} < \infty. \quad (3.4)$$

Solution estimate

Theorem [LHD, Kloeden (2023)]: Consider the RDE

$$d\hat{y}_t = f(\hat{y}_t)dt + g(\hat{y}_t)dX_t, \hat{y}_a \in \mathbb{R}^d, t \in [a, b] \quad (3.5)$$

where X is a stochastic process with stationary increment which almost all realizations x are of local Hölder continuous with exponent in $(1/3, 1/2]$ such that x is truly rough.

If in addition to $(\mathbf{H}_f)$, $(\mathbf{H}_g)$, f and D^2g are local Lipchitz, then there exists a generic random variable $\hat{\xi}(\hat{y}_a, \mathbf{x}, [a, b])$ such that the following estimate holds

$$\begin{aligned} & \sup_{0 \leq k \leq m} \|\hat{y}(t_k, y_a, x) - y_{t_k}\| \\ & \leq \hat{\xi}(\hat{y}_a, \mathbf{x}, [a, b])(|\Pi[a, b]| + \|\mathbf{x}\|_{p, [a, b], |\Pi[a, b]|})^{3-p}(b - a + \|\mathbf{x}\|_{p, [a, b]}^p). \end{aligned} \quad (3.6)$$

In particular,

$$\lim_{|\Pi[a, b]| \rightarrow 0} \sup_{0 \leq k \leq m} \|\hat{y}(t_k, \hat{y}_a, x) - y_{t_k}\| = 0. \quad (3.7)$$

Theorem [NDCong-LHD-PTHong (2023)]: give a direct proof using discrete rough paths.

Fitzhugh-Nagumo neuro-system

The deterministic FHN neuron model is

$$\begin{cases} dv_t &= v_t - \frac{v_t^3}{3} - w_t + I \\ dw_t &= \varepsilon(v_t - \mu w_t + J) \end{cases} \quad (4.1)$$

It is important to note that although our vector field f is dissipative, it does not satisfy the linear growth in the perpendicular direction

$$\left\| f(y) - \frac{\langle f(y), y \rangle}{\|y\|^2} y \right\| \leq C_f (1 + \|y\|), \quad \forall y \neq 0 \quad (4.2)$$

for a certain constant C_f . Instead, we introduce a different Lyapunov function

$$V(y) := \left(1 + v^4 + \frac{6}{\varepsilon\mu} w^2 \right)^{\frac{1}{4}}. \quad (4.3)$$

Theorem [LHD-J.Jost (2023)]: Under the assumptions $(\mathbf{H}_g^b)$, $(\mathbf{H}_X)$, there exists a solution of stochastic FHN system on any interval $[0, T]$. Moreover, there exist constants $\delta > 0$ such that for any $\lambda \in (0, 1)$ small enough, there exist a constant $K_\lambda > 0$ such that the following estimate holds for all

$$V(y_t) - K_\lambda \leq e^{-\delta t} \left[V(y_0) - K_\lambda \right] + L_V \lambda \int_0^t e^{-\delta s} dN^-\left(\frac{\lambda}{16C_p C_g}, \mathbf{x}, [t-s, t]\right), \quad \forall t \in [0, T]. \quad (4.4)$$

Consequently, there exists a global pullback attractor $\mathcal{A}$.

References

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THANK YOU!